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Join our lab:



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Research

- Quantum computing for machine learning
- Machine learning for quantum computing
- Hardware accelerator
- Functional programming

Paper: journal [14], conference [28], book [2] (September 2025)

Fund: NAIST Granite, Unitary Fund, ...

Reviewer: SOICT, SoftwareX, Quantum Information Processing, ...

See more: <https://aiseqlab.github.io/demo/>



Vu Tuan Hai

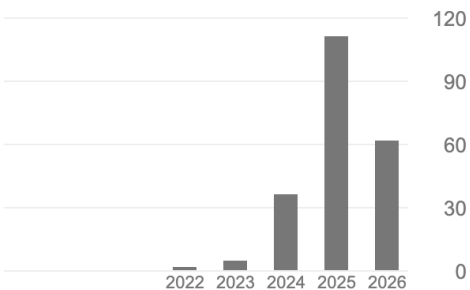
Other names >

University of Information Technology, Vietnam National University HCM city
Verified email at uit.edu.vn - [Homepage](#)
[quantum computing](#) [deep learning](#) [emulator](#) [cryptography](#)

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Based on funding mandates

☐	TITLE	CITED BY	YEAR
☐	On the Interplay Between Noise, Bell Violation, and Cascade Error Correction in Device-Independent Quantum Key Distribution NDH Duy, NT Dong, VT Hai, LVT Duong, N Van Tinh arXiv preprint arXiv:2604.22232		2026
☐	Towards Automated Selection of Quantum Encoding Circuits via Meta-Learning DD Tung, NQ Chuong, VT Hai, LB Ho, LN Tran arXiv preprint arXiv:2604.19076		2026
☐	UniCrypt-Ultra: A 64-bit Quad-Core Universal Cryptographic Engine for Multi-mode Cryptographic Processing in Blockchain IoT and Society 5.0 VTD Le, HL Pham, TH Vu, VD Tran, Y Nakashima IEEE Micro		2026
☐	Crypto-RV: High-Efficiency FPGA-Based RISC-V Cryptographic Co-Processor for IoT Security AK Pham, VT Vo, VTD Le, TH Vu, HL Pham, VT Nguyen, Y Nakashima arXiv preprint arXiv:2602.04415		2026

1/2019

5/2021

2/2022

6/2023

11/2023

10/2025

Join ICIP lab
as newbie

Graduated
1st journal
paper published

TA at SE, UIT
Master student

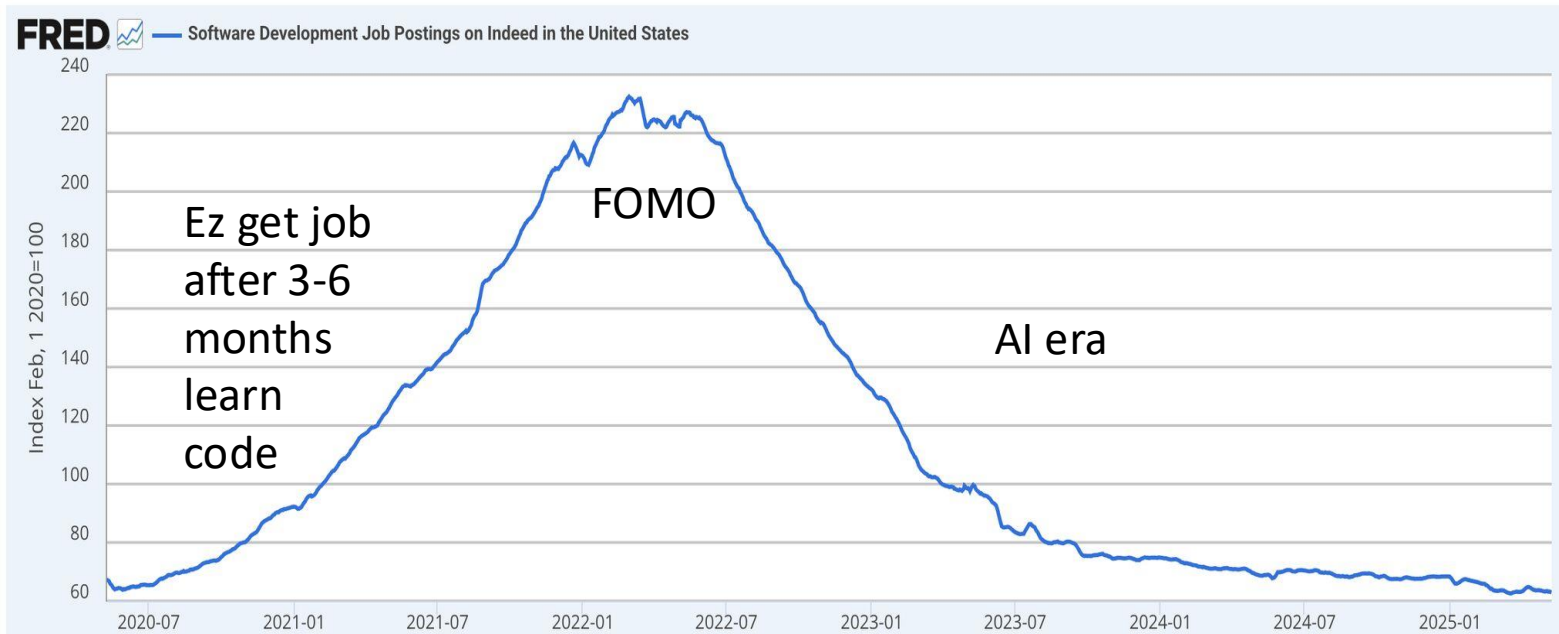
Graduated
3rd journal paper
published

Got PhD
scholarship at
NAIST, Japan

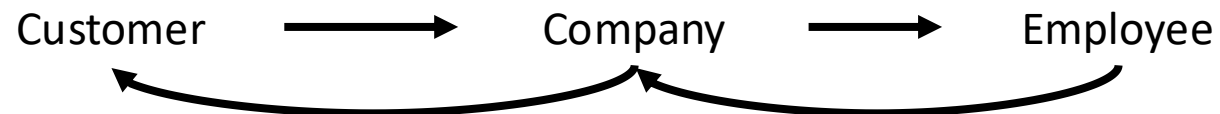
Graduated
Back to UIT

Many works
were published!

Why research? (1/3)



Outsource



Low risk, low return

R&D



High risk, high return

Job market is
changing now
"SE job declining"



Why research? (1/3)

Giáo dục > Tin tức

Thứ năm, 2/4/2026, 06:05 (GMT+7)

Chuẩn 'đại học' mới yêu cầu 60% giảng viên là tiến sĩ

Các "trường đại học" phải có tỷ lệ giảng viên là tiến sĩ tối thiểu 60%, tỷ lệ nguồn thu từ khoa học công nghệ đạt 20% mới được công nhận là "đại học", khó hơn trước đây.



Viettel High Tech

Chuyên gia Công nghệ lượng tử-

Hanoi Capital Region · 10 months ago · **0 applicants**

Company review time is typically 1 week

On-site

Full-time



No longer accepting applications



Endava

Junior Web Developer ✓

Ho Chi Minh City, Ho Chi Minh City, Vietnam · Reposted 2 weeks ago · Over 100 people clicked apply

Promoted by hirer · Responses managed off LinkedIn

Hybrid

Full-time

Apply

Save

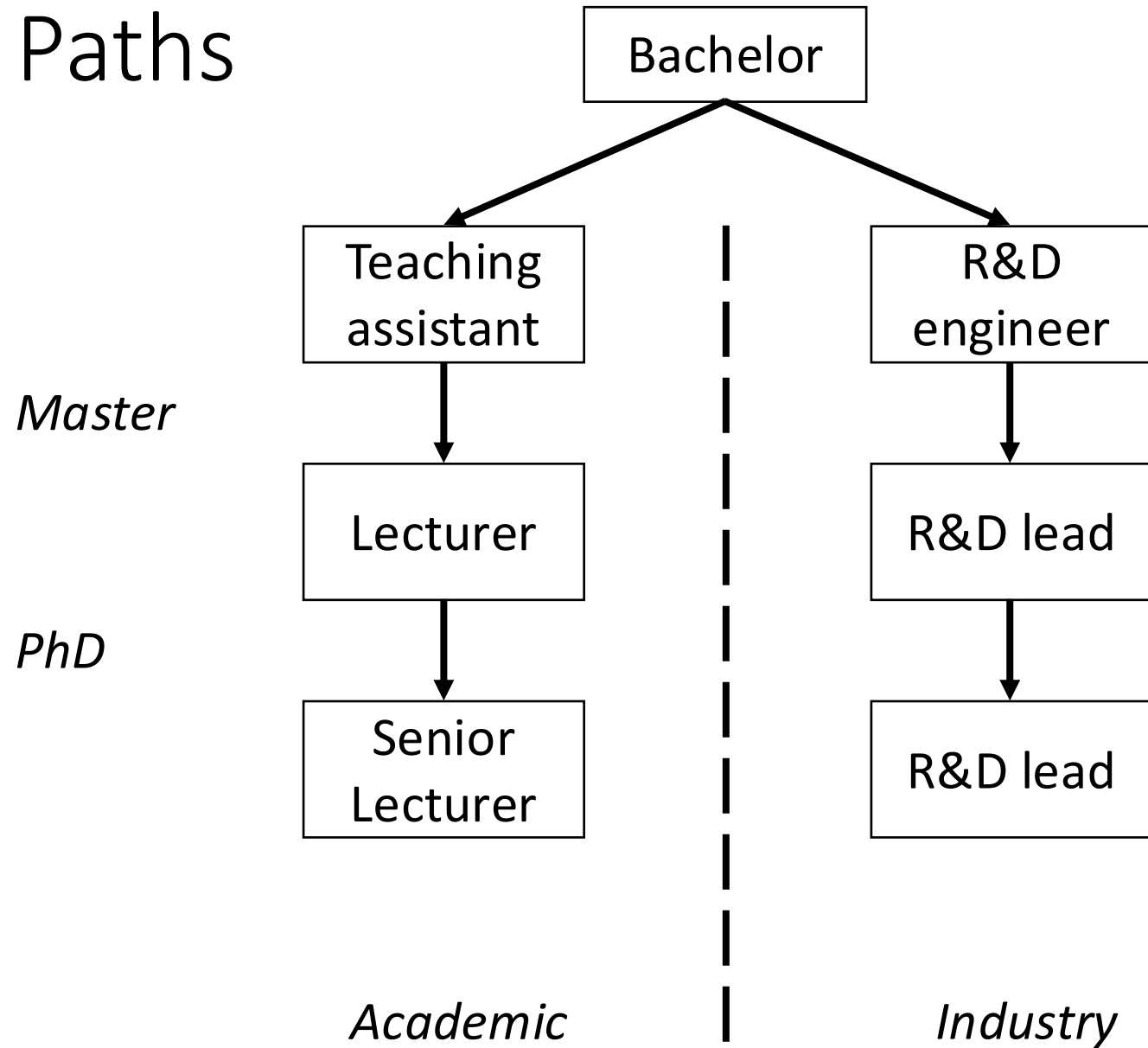
Why research? (2/3): Research → Many skills

	Coding	Coding + Research
Ez?	Ez	Hard
Self-learning	Possible	Need supervisor + lab equipment
Required skills	Coding	Problem solving Visualization Presenting skill Teamwork Coding ...
Spending time + health	Short, only take 3-6 months	Long, few years

Why research? (3/3): Job market

	Coder	Researcher
Money	Average	Much better (R&D, lecturer, ...)
Credits	Project	Project + Paper + Patent + Relationship
Number of positions	Decrease	Increase

Paths



Simple guideline

1. Choose your interested research aspects
2. Find a suitable supervisor
3. Keep going on!

Detail steps

3. Keep going on

- (a) Balancing between researching and studying
- (b) Group or solo?
- (c) Research method
- (d) Research tool
- (e) Research outcome
- (f) Use the outcome for: scholarship, job, ...